Dual-Band Terahertz Metamaterial Absorber with Enhanced Sensing Capabilities

Tejashree S
Department of ECE
Sai Vidya Institute of Technology
Rajanukunte, India
tejashree.s@saividya.ac.in

Chaya B M
Department of ECE
Sai Vidya Institute of Technology
Rajanukunte, India
chaya.bm@saividya.ac.in

Likith S
Department of ECE
Sai Vidya Institute of Technology
Rajanukunte, India
likiths.22ec@saividya.ac.in

Chethan R
Department of ECE
Sai Vidya Institute of Technology
Rajanukunte, India
chethanr.22ec@saividya.ac.in

***Abstract*—The growing demand for highly sensitive and compact biosensors has driven significant interest in terahertz metamaterial absorbers for biomedical diagnostics and refractive index sensing applications. In this work, a dual-band terahertz metamaterial absorber is proposed and numerically analyzed for high-performance label-free biosensing. The designed structure consists of a square metallic ring integrated with a periodically arranged Z-shaped resonator. The proposed absorber exhibits two strong resonance frequencies at 0.7421 THz and 0.9061 THz with absorption rates exceeding 99.2% and 97.79% , respectively, under both TE and TM modes at normal incidence. Narrow full width at half maximum (FWHM) values of 0.0925 THz and 0.1502 THz result in high Q-factors of 8.0203 and 6.041. The absorber also demonstrates stable performance against variations in polarization and incident angles, ensuring robust operation. Parametric investigations confirm the tunable and adaptable behavior of the proposed design. Furthermore, the sensor achieves average sensitivities ranging from 2.797 to 2.801 THz/RIU and Figures of Merit (FOMs) ranging from 18.6305 RIU^{-1} to 30.2824 RIU^{-1}, making it a promising candidate for advanced terahertz biosensing and biomedical diagnostic applications.**

***Index Terms*—Metamaterial, Absorber, Terahertz, Quality factor, High sensitivity, FOM, Dual Band.**

I. Introduction

Electromagnetic (EM) metamaterials are artificially engineered composite materials designed to exhibit extraordinary electromagnetic properties that are rarely found in naturally occurring materials [1]. Due to their unique ability to manipulate electromagnetic waves in unconventional ways, metamaterials have attracted significant attention in recent years. Researchers have extensively explored these materials for applications such as negative refractive index media, super lenses, compact wireless devices [2], and highly efficient absorbers [3]. Their exceptional light-controlling capability has opened new possibilities in modern electromagnetic and photonic technologies.

Artificially designed electromagnetic absorbers based on metamaterials have demonstrated promising performance in various fields, including solar energy harvesting, spectral imaging, non-destructive sensing, stealth technology, wireless communication, and emissivity control. Among the different metamaterial configurations, absorbers composed of specially designed electric and magnetic resonators have gained considerable research interest. Most perfect metamaterial absorbers are typically constructed using patterned metallic structures deposited on dielectric substrates [4]. Owing to their near-perfect absorption characteristics, these absorbers have found wide-ranging applications in spectroscopic detection, thermal imaging systems and advanced optical devices.

With the rapid advancement of terahertz (THz) based technologies, the design and development of perfect absorbers have emerged as a significant area of research in recent years [5]. Since the introduction of the first perfect absorber by N. I. Landy et al. [6], the development of metamaterial perfect absorbers has attracted widespread research interest. Subsequently, in 2008, H. Tao et al. [7] demonstrated the first terahertz (THz) metamaterial absorber, achieving nearly 98% absorption at 1.12 THz through both numerical simulations and experimental validation.

In this paper, a terahertz (THz) metamaterial absorber with near-perfect absorption characteristics is proposed and investigated. The designed structure exhibits dual-band absorption with high absorption efficiency at both resonant frequencies. The absorber demonstrates excellent performance under both transverse electric (TE) and transverse magnetic (TM) polarization modes. In addition, the proposed metamaterial absorber shows strong stability against variations in incident and polarization angles for both TE and TM modes, enabling consistent dual-band absorption over a wide angular range.

The paper is organized as follows. The first section presents the structural design and configuration of the proposed metamaterial absorber, followed by a detailed discussion of the simulation results. Subsequently, the electric field,

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magnetic field, and surface current distribution analyses are presented to explain the origin and behavior of the absorption resonances. A comprehensive parametric study is then carried out to evaluate the effect of structural parameters and to justify the selected design dimensions.

II. Structure and Design

The physical dimension of the metamaterial absorber unit cell is shown in Fig. 1, which consists of a single Z-shaped film inside a Ring-shaped outer design, both made up of copper. The dimensions of the proposed metamaterial absorber

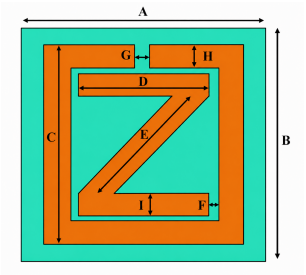


Fig. 1. Metamaterial Absorber Unit Cell.

were carefully selected and optimized to achieve efficient dual-band absorption in the terahertz region. The overall size of the unit cell is defined by parameters A and B, both having dimensions of 50µm. The outer square resonator is designed with a side length C of 40µm. Inside the structure, the horizontal section represented by parameter D measures 26.5µm, while the diagonal Z-shaped section E has a length of 31.6µm. The gaps between different resonator sections, denoted by parameters F and G, are maintained at 2µm to ensure proper electromagnetic coupling. In addition, the strip widths represented by parameters H and I are both fixed at 5µm. All dimensions are expressed in micrometers (µm) and were optimized to obtain strong absorption characteristics along with stable electromagnetic performance.

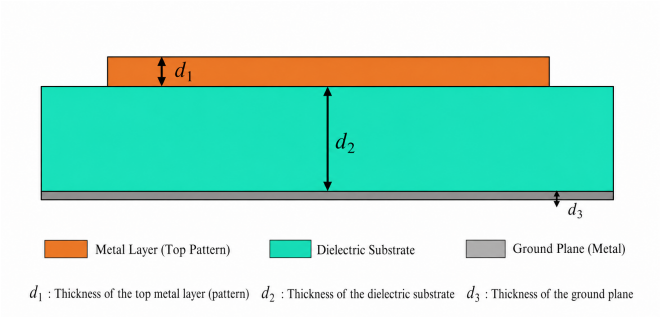


Fig. 2. Cross-sectional view of Metamaterial Absorber.

Fig 2 presents the cross-sectional view of the meta-surface absorber, showing the metallic resonator layer placed over a dielectric substrate and backed by a ground layer. The parameters d1, d2, and d3 represent the thicknesses of the resonator, substrate, and ground layer, respectively, which influence the absorption performance. Here the geometric parameters are d1 = 0.05µm thickness of the lower metal layer, d2 = 10µm thickness of the SiO2 substrate, d3 = 2.5µm is the metal layer thickness, The ”Z” diagonal angle with respect to the x-axis is of 133.5 degrees. P = Px = Py =50[µm] Px and Py represent the period size in the x- and y-direction, respectively. In this work, the finite element method [FEM] is applied to investigate the resonant behavior of the perfect absorber. During the simulation, the Boundary conditions in the X, Y directions are set as Periodic while Perfectly Matched Layers (PMLs) are applied along the z-direction.

III. Modelling of Metamaterial Absorber

The proposed absorber was simulated using COMSOL Multiphysics to design and analyze the proposed MMA, emphasizing its structural features and performance capabilities. The FEM has been employed to perform all the simulations. In the simulations, the unit-cell periodic boundary condition is set for the proposed periodic array. The mesh size needs to satisfy the convergence and accuracy requirements. The plane waves are incident to illuminate the proposed absorber. The frequency-dependent absorption rate A(ω) can be computed as:

$$A(\omega) = 1 - R(\omega) - T(\omega) \tag{1}$$

where R(ω) is expressed as S11(ω) representing the reflection coefficient and T(ω) is equivalent to S21(ω), representing the transmission coefficient. Since the gold ground plane on the back of the silicon wafer being assumed as a fully reflecting surface, the transmission coefficient T(ω) can be considered as zero. Hence, the absorption rate can be simplified as:

$$A(\omega) = 1 - R(\omega) = 1 - |S_{11}|^2 \tag{2}$$

A single bottom layer of SiO2 is used with a thickness of 0.05um as a lossy medium with conductivity independent of frequency $\sigma = 6.30 \times 10^7$ S/m Metamaterials are specially engineered materials that possess distinct properties, with a permittivity (ϵ) that is lower than the free space permittivity (ε_0), and a free space permeability (μ_0) .The impedance (Z) of the ideal absorber should be closely matched to the free space impedance to reduce the reflection from the surface [8].

$$Z_{in}(\omega) = Z_0 \sqrt{\frac{(1+S_{11})^2 - S_{21}^2}{(1-S_{11})^2 - S_{21}^2}} \tag{3}$$

where S21 is the transmission coefficient, and ω is the angular frequency.

The performance of any sensor is typically evaluated using three key metrics: the quality factor (Q), sensitivity (S), and figure of merit (FOM) [9]. The quality factor reflects the resonant behavior of the sensor and is directly linked to its resolution and sensitivity. A higher Q value indicates lower dielectric losses, resulting in a sharper and narrower resonance peak, which enhances both resolution and sensitivity. It is mathematically defined as.

$$Q = \frac{f_{res}}{FWHM} \tag{4}$$

denotes the full width at half maximum of the resonance peak. Sensitivity (S) quantifies the shift in frequency of the resonant peak in response to a unit change in the refractive index(RI) of the analyte. It is expressed as:

$$S = \frac{\Delta f}{\Delta n} \tag{5}$$

The figure of merit (FOM) combines sensitivity and spectral resolution into a single dimensionless parameter, defined as:

$$FOM = \frac{S}{FWHM} \tag{6}$$

In refractive index-based sensing, when the refractive index change is fixed, a larger shift in resonant frequency indicates higher sensitivity. Sensitivity is typically measured in units of GHz or THz/RIU, representing the frequency response per unit change in refractive index [10,11].
The simulation of the symmetrical absorber design is based on the full vectorial finite element approach utilizing COMSOL Multiphysics.

IV. Simulations and Discussion

The proposed structure is obtained by integrating two individual designs. The first configuration, referred to as absorber 1, consists of a basic single square split-ring resonator (SRR). This design exhibits a resonant frequency of 0.7329 THz with an absorption efficiency of 99.2% at that frequency. The second configuration, referred to as Absorber 2, employs a Z-shaped unit cell, which demonstrates a resonant frequency at 0.9359 THz and achieves an absorption of 96.7%.
By combining these two configurations, the final structure produces a dual-band response with resonant frequencies occurring at 0.7421 THz and 0.9016 THz, where absorption values of 99.2% and 97.79% are obtained, respectively. The unit cell of the structure consists of copper, which has an intrinsic electrical conductivity of $\sigma = 6.30 \times 10^7$ S/m with a thickness of 2.05 μm. To enhance reflection, a silver layer is employed as the metallic ground plane towards the bottom, while a silicon substrate with a relative permittivity of ϵ = 10.8 is placed above it.

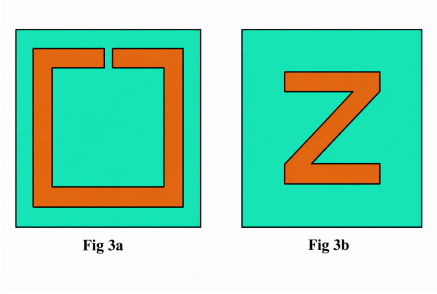


Fig. 3. Depict Design 1 and Design 2.

Figure 3 (a) and (b) respectively depict Design 1 and Design 2.

Figure 4 and 5 depicts the development of the proposed structure's absorption response. The procedure for designing

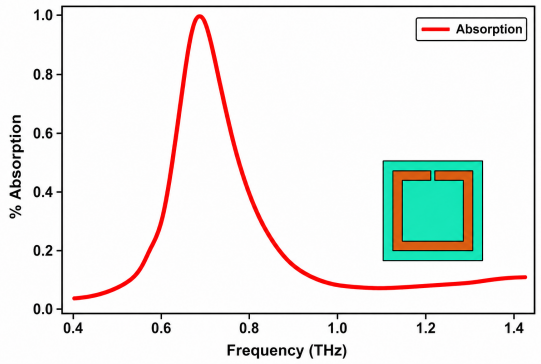


Fig. 4. Absorbance of Design 1.

Absorber 1 and Absorber 2 is identical, with the exception of the unit cell shape.

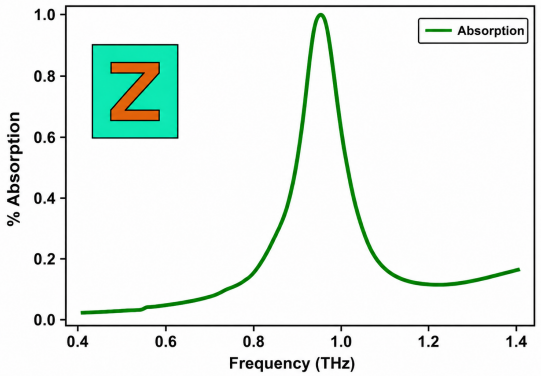


Fig. 5. Absorbance of Design 2.

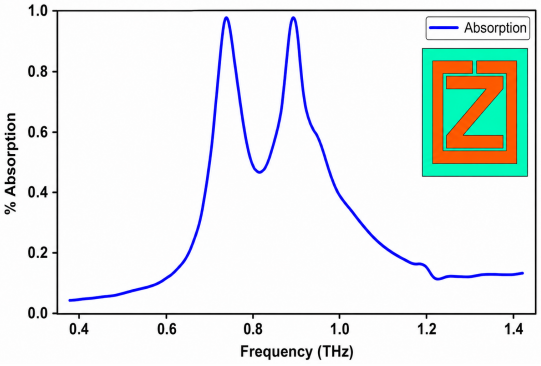


Fig. 6. Combined Absorbance of Design 1 and 2.

Figure 6 depicts the proposed design's absorption response as a combination of Design 1 and Design 2.

Design 1 has resonances at 0.7239 THz, while Design 2 has resonances at 0.9359 THz. Both the proposed design's resonant frequencies undergo a red shift as a result of the interaction between Design 1 and Design 2, resulting in two distinct absorption peaks at 0.7329 THz and 0.9539 THz. The coupling effect at these resonant frequencies between Design 1 and Design 2 makes it easier to absorb nearly perfectly.

The proposed design exhibits narrow bandwidths at both resonant frequencies of 0.7421 THz and 0.9016 THz.

In Field Analysis, the operating mechanism of the proposed MMA is analyzed through the distributions of the electric (E) field, magnetic (H) field, and the surface current. Figure 7 illustrates the E-field distribution within the unit cell array, while Figure 8 presents the corresponding H-field distributions at the resonance frequencies of 0.7421 THz and 0.9016 THz.

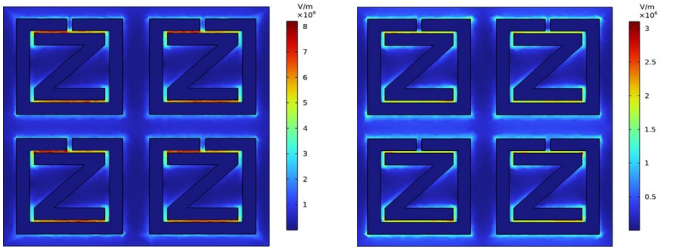


Fig. 7. E Field at 0.7421 Thz and 0.9061 Thz.

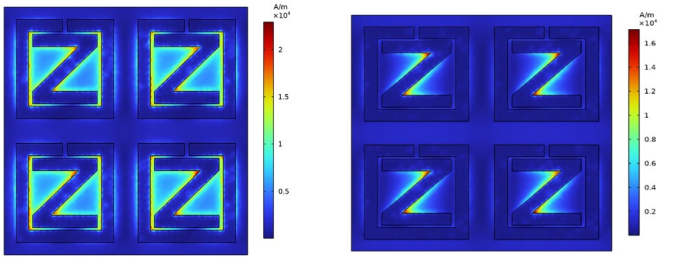


Fig. 8. H Field at 0.7421 Thz and 0.9061 Thz.

V. Comparative Analysis of Metallic Materials

The performance of a metamaterial absorber is predominantly governed by the choice of metallic material, as its intrinsic conductivity and loss characteristics directly influence resonance frequency, absorbance, quality factor (Q-factor), and FOM.

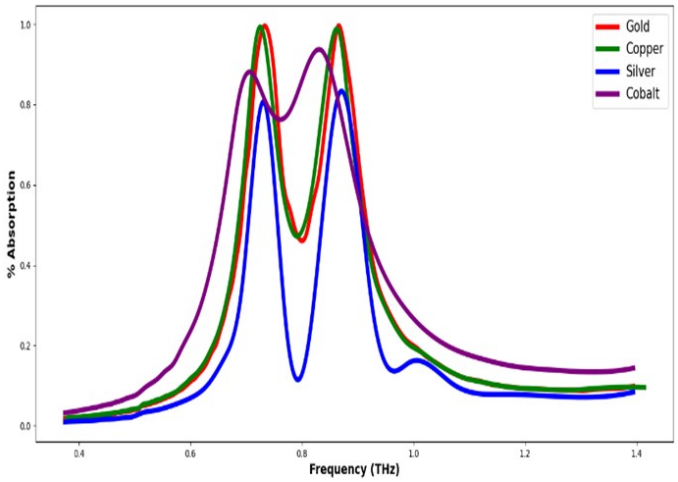


Fig. 9. Combined plot of various materials for their absorbance.

TABLE I
Comparison of the Proposed Absorber with Previous Works

Parameter	[15]	[16]	This Work
Absorber Shape	Array of Rings	Multilayer Structure	Ring with Z-shaped Resonator
Resonant Frequency	4.5–5.6 THz	4.5–5.6 THz	0.7421–0.9061 THz
Maximum Absorbance	99%	>95%	98–99.5%
Quality Factor (Q)	8.948	5.359	8.0203
Sensitivity	3.880 THz/RIU	429 GHz/RIU	2.801 THz/RIU

As shown in Fig. 9, gold provides near-unity absorbance ($\sim$99.5%) with stable Q-factors; however, its high cost limits large-scale applications. Silver offers the sharpest resonances ($Q \approx 9.4$) owing to its low intrinsic losses, although oxidation adversely affects its long-term reliability. Copper achieves high absorbance (97–99%) with moderate Q-factors and excellent scalability, whereas cobalt exhibits low Q-factors and broad resonances, making it less suitable for biosensing applications.

Among the hybrid structures, the copper–silver configuration demonstrates the best overall performance, achieving the highest Q-factor (8.02) and a high figure of merit (FOM) of $5 \times 10^1\ \mathrm{RIU}^{-1}$. This improvement arises from synergistic plasmonic coupling, where silver enhances the resonance sharpness while copper provides structural stability and cost-effectiveness.

VI. Conclusion

A dual-band terahertz metamaterial absorber incorporating a Z-shaped resonator has been proposed and numerically investigated for label-free biosensing applications. The proposed absorber achieves near-unity absorption of 99.2% and 97.79% at resonance frequencies of 0.7421 THz and 0.9061 THz, respectively, with corresponding Q-factors of 8.0203 and 6.041. The structure exhibits stable performance under both TE and TM polarizations and maintains excellent absorption characteristics over a wide range of polarization and incident angles. Furthermore, the proposed sensor demonstrates a sensitivity of 2.801 THz/RIU and a maximum figure of merit of 30.2824 RIU^{-1}, indicating its suitability for high-performance refractive-index sensing.

A comparative study of different metallic materials further shows that the hybrid copper–silver configuration provides an effective balance between sensing performance, fabrication feasibility, and cost. These characteristics make the proposed absorber a promising candidate for terahertz biosensing, biomedical diagnostics, and other advanced sensing applications.

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